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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

MINI PROJECT REPORT

EMPLOYEE ATTRITION ANALYSIS

Exploratory Data Analysis, Visualization and Data Preprocessing

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Under the Guidance of

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CERTIFICATE

This is to certify that the mini project report titled **"Employee Attrition Analysis"** is submitted by **S THRISHA** and **M SRI SREYA**, from the Department of Artificial Intelligence and Data Science, SRM TRP Engineering College. The project focuses on analyzing employee data to understand the factors that lead to employee attrition. It uses data preprocessing, exploratory data analysis, and visualization to identify patterns and trends related to employee turnover and gain insights into employee retention.

The report is submitted as part of the academic mini project work under the guidance of **V. Ezhil Swanly**.

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Finally, I am deeply grateful to my family and friends for their constant encouragement, motivation, and support throughout the completion of this project.

This project has helped me enhance my knowledge of data analysis and understand how data-driven insights can be used to identify the factors influencing employee attrition and support better human resource management decisions.

SRI SREYA M

THRISHA S

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ABSTRACT

Employee attrition refers to employees leaving an organization during a particular period. Understanding employee attrition is important for studying workforce stability and identifying employee-related patterns that may require attention.

This project, "Employee Attrition Analysis," applies data preprocessing, exploratory data analysis (EDA), descriptive statistics, and visualization to an employee dataset. Depending on the columns available in the dataset, attributes may include age, department, job role, job satisfaction, monthly income, years at the company, overtime, work-life balance, and an attrition label.

Python, Pandas, Matplotlib, and Seaborn can be used to inspect the dataset, identify missing values and duplicate records, summarize numeric and categorical variables, and create charts such as attrition distribution, attrition by department, age or tenure distribution, overtime versus attrition, and a correlation heatmap for numeric features.

The purpose is to discover descriptive patterns and communicate them clearly. Exact record counts, missing-value totals, percentages, and chart-based findings must be filled in after running the analysis on the project's actual dataset. This is an exploratory educational project; associations in the data do not prove causes, and no predictive model should be claimed unless one is trained and evaluated separately.

CHAPTER 1: INTRODUCTION

1.1 Background

- Organizations depend on employees' skills, experience, and performance to achieve their goals. When employees leave, organizations may need to recruit and train replacements, which can affect continuity and incur additional costs.
- Employee datasets can be examined to understand how attrition is distributed across departments, job roles, age groups, tenure, job satisfaction, overtime, and other available attributes. Exploratory Data Analysis (EDA) helps convert raw employee records into summaries and visual patterns that are easier to interpret.

1.2 Problem Statement

- Raw employee data may contain missing values, duplicate records, inconsistent categories, and a mixture of numeric and categorical variables. Without systematic inspection, it can be difficult to understand the distribution of attrition or compare employee groups.

- This project addresses the problem by preparing the available data, summarizing its characteristics, and visualizing employee attrition patterns without assuming that any single factor causes an employee to leave

1.3 Objectives

- To inspect the employee dataset and identify its dimensions, data types, missing values, and duplicate records.
- To clean and prepare the available data for reliable descriptive analysis.
- To summarize employee attributes using descriptive statistics and frequency counts.
- To visualize attrition distribution and compare attrition across relevant employee groups.
- To explore relationships between selected numeric variables using correlation analysis where appropriate.
- To document findings, limitations, and possible directions for future analysis.

1.4 Scope

- The project focuses on exploratory analysis of the employee dataset provided for academic use. The analysis is limited to the available fields and records.
- It is intended to describe patterns in the dataset, not to make decisions about individual employees, establish causal relationships, or provide a validated prediction system.

1.5 Expected Outcome

- The expected outputs are an inspected and cleaned dataset, a reproducible Python notebook, descriptive summaries, relevant visualizations, and a written interpretation of the observed patterns.
- The actual outputs and numerical findings should be updated after the analysis is run on the chosen dataset.

CHAPTER 2: DATASET DESCRIPTION

2.1 Dataset Source

- The project uses the employee attrition dataset selected by the project team. The dataset is used to inspect employee information and analyze patterns associated with attrition.
- The dataset used for the Employee Attrition Analysis project contains employee-related information that can be used to study workforce patterns and employee turnover. The dataset may be obtained from publicly available datasets, such as IBM HR Analytics Employee Attrition and Performance, available through platforms like Kaggle.

2.2 Dataset Overview

- The Employee Attrition Analysis dataset contains information about employees and their workplace-related characteristics. It provides a structured view of employee details that can be analyzed to understand workforce distribution and attrition patterns.
- The dataset may include attributes such as age, department, job role, monthly income, job satisfaction, years at the company, overtime, and attrition status, depending on the dataset selected.

2.3 Attribute Description

- The following table contains common examples of attributes that may be present in an employee attrition dataset. Only the attributes available in the actual dataset should be included in the final report.

Attribute	Description	Role
Age	Employee age in years	Numeric feature
Department	Organizational department	Categorical feature
Job Role	Employee job role	Categorical feature
Monthly Income	Monthly income value	Numeric feature
Job Satisfaction	Recorded job satisfaction rating	Ordinal/categorical feature
YearsAtCompany	Years employed at the organization	Numeric feature
Overtime	Whether overtime is recorded	Categorical feature
Attrition	Whether the employee left, if present	Outcome/target label

2.4 Dataset Preparation

- Column names, category labels, units, and coding schemes should be reviewed before analysis. If a category such as Attrition is encoded numerically, its mapping must be documented.
- For example, the meaning of each numeric code should be verified from the dataset documentation rather than assumed. Category codes should not be treated as continuous measurements unless the dataset definition supports that use.

CHAPTER 3: METHODOLOGY

3.1 Project Workflow

1. Load the raw employee dataset into Python using Pandas.
2. Inspect the first and last records, shape, column names, data types, and summary information.
3. Measure missing values and duplicate records and review inconsistent category labels.
4. Prepare a cleaned analysis copy and document every change made.
5. Generate descriptive statistics and counts for numeric and categorical fields.
6. Create visualizations to examine attrition and compare groups using the available fields.
7. Interpret results cautiously, note limitations, and prepare the report and charts.

3.2 Tools and Technologies

- Python: Used as the primary programming language for data analysis.
- Pandas: Used for loading, cleaning, grouping, and summarizing tabular data.
- Matplotlib: Used to create charts and graphical representations of the data.
- Seaborn: Used to create statistical visualizations such as count plots, box plots, and correlation heatmaps.
- Microsoft Excel: Can be used to inspect and organize raw and cleaned datasets.
- Google Colab / Jupyter Notebook: Used to execute Python code and document the analysis process.

3.3 Analysis Approach

- The analysis combines data quality checks, descriptive statistics, group-wise comparisons, and visual exploration.
- If an Attrition field is available, its frequency and percentage can be calculated and compared across selected categories such as department, job role, overtime, or tenure groups.
- Correlation analysis should be applied only to suitable numeric variables, with categorical encoding and interpretation documented.

CHAPTER 4: DATA PREPROCESSING

4.1 Data Inspection

- The dataset is inspected to determine its number of rows and columns, variable types, missing-value counts, and descriptive statistics.
- The inspection also helps identify whether columns contain unexpected values or inconsistent formatting. The actual outputs from the analysis notebook should be included in this section.

4.2 Cleaning and Preparation

- Missing values should be handled according to the meaning of each field and the project requirements. Possible approaches include documenting missingness, removing records only when justified, or using an appropriate imputation method.
- Duplicate records should be investigated before removal. Category spelling and whitespace can be standardized where needed.
- The report must state the exact steps actually performed. The dataset should not be described as completely cleaned unless the relevant checks confirm it.

4.3 Categorical Encoding

- Categorical variables can be summarized directly by category. Encoding may be needed for certain computations or a future machine-learning stage.
- Any encoding should preserve a clear mapping between codes and labels. Nominal categories should not be interpreted as ordered numeric values simply because they have been assigned numbers.

4.4 Feature Review

- The features reviewed depend on the columns available in the employee dataset. Examples include Age, Department, Job Role, Monthly Income, Job Satisfaction, YearsAtCompany, Overtime, and Attrition.
- Only the fields that exist in the selected dataset should be included. Any fields removed from the analysis should be documented along with the reason for removal.

4.5 Clean Dataset Summary

After preprocessing, report the final number of records and columns, missing-value totals, duplicate handling, and any changes made to data types or category labels.

- Total Records: Number of employee records available after cleaning.
- Total Attributes: Number of columns retained for analysis.
- Missing Values: Missing entries remaining after preprocessing.
- Duplicate Records: Duplicate entries identified and handled.
- Data Consistency: Data types and values checked for consistency.
- Analysis Readiness: Dataset prepared for exploratory data analysis and visualization.

CHAPTER 5: EXPLORATORY DATA ANALYSIS

5.1 Dataset Summary

- The dataset summary provides an overview of the employee records and their attributes.
- For numeric fields, measures such as mean, median, standard deviation, minimum, and maximum can be calculated where meaningful. For categorical fields, frequency counts and percentages can be used.
- The summary also includes the dataset size, variable types, missing values, and duplicate count.

5.2 Attrition Distribution

- A count plot or bar chart can be used to display the number of employees in each Attrition category, if the field exists.
- A percentage view can help compare the proportions of employees who left and those who stayed.
- The actual counts and percentages must be inserted after running the analysis. If the dataset does not contain an Attrition label, direct attrition-rate analysis is not possible with the available data.

5.4 Age and Tenure Analysis

- Histograms or box plots can be used to explore the distributions of Age, YearsAtCompany, or other relevant numeric attributes.
- If an Attrition field is available, these distributions may be compared between employees who left and those who stayed.
- The observations should be based on the actual plots generated from the dataset.

5.5 Overtime, Satisfaction, and Other Factors

- Where these fields exist, attrition can be compared across Overtime categories or Job Satisfaction levels.
- A grouped bar chart or normalized percentage chart may be useful for this comparison.
- The meaning and coding of each variable should be checked before interpretation. The chart can reveal an association in the sample but cannot establish a causal relationship.

5.6 Correlation Analysis

- A correlation heatmap can summarize pairwise relationships among appropriate numeric variables.
- Correlation measures association and does not establish causation. Encoded nominal categories should generally not be included as if their numeric codes were continuous measurements.
- The variables used and the calculated coefficients should be reported only after performing the actual analysis.

5.7 Visualizations

Figure 5.1:

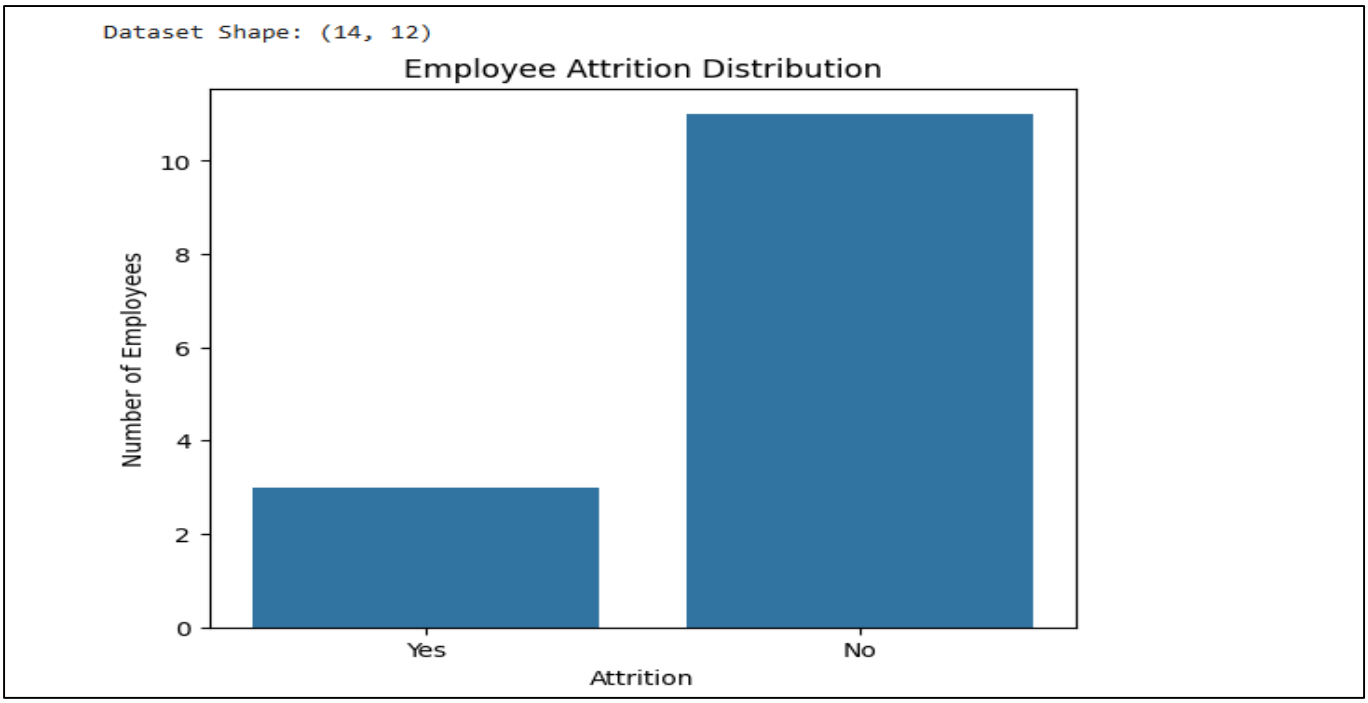


Figure 5.2:

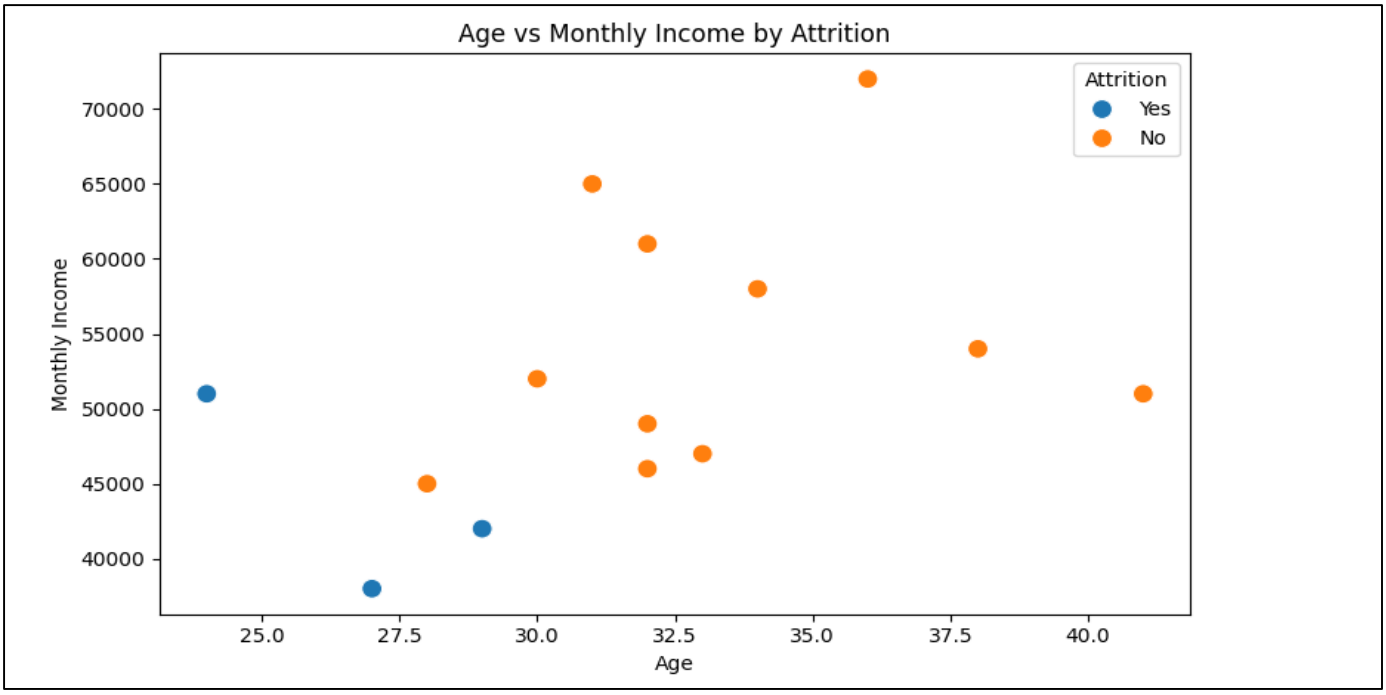
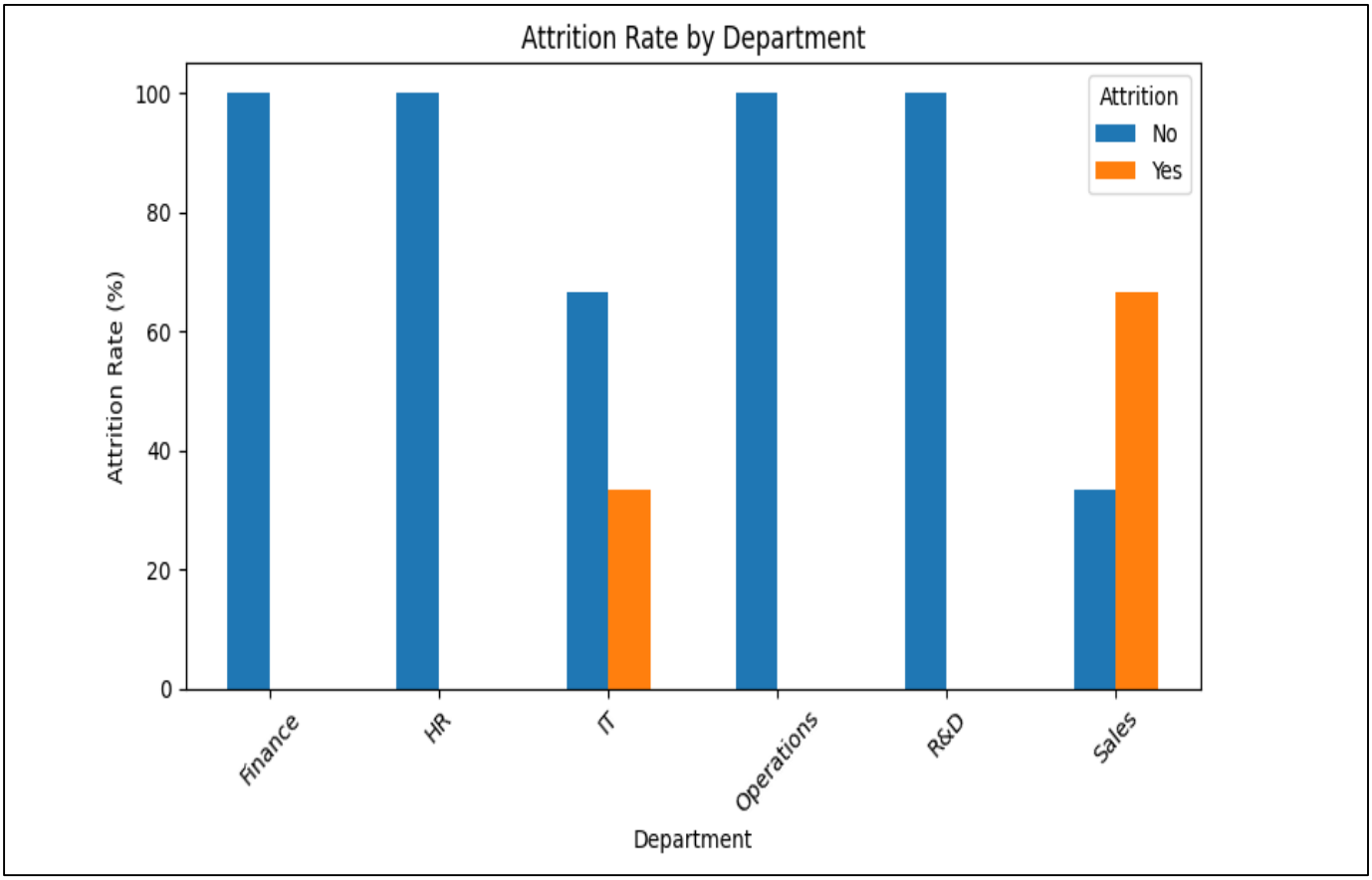


Figure 5.3:



CHAPTER 6: RESULTS AND INTERPRETATION

6.1 Attrition Overview

- Employee attrition refers to the number of employees leaving an organization over a specific period. In this project, attrition analysis is used to understand employee turnover patterns and identify the factors that may influence employees to leave the company. The dataset is examined to compare employees who have left the organization with those who continue to work there.

6.2 Group Comparisons

- The analysis can summarize the differences observed between available groups, such as department, job role, overtime status, or satisfaction level.
- The relevant charts should indicate whether the comparison uses counts or percentages.

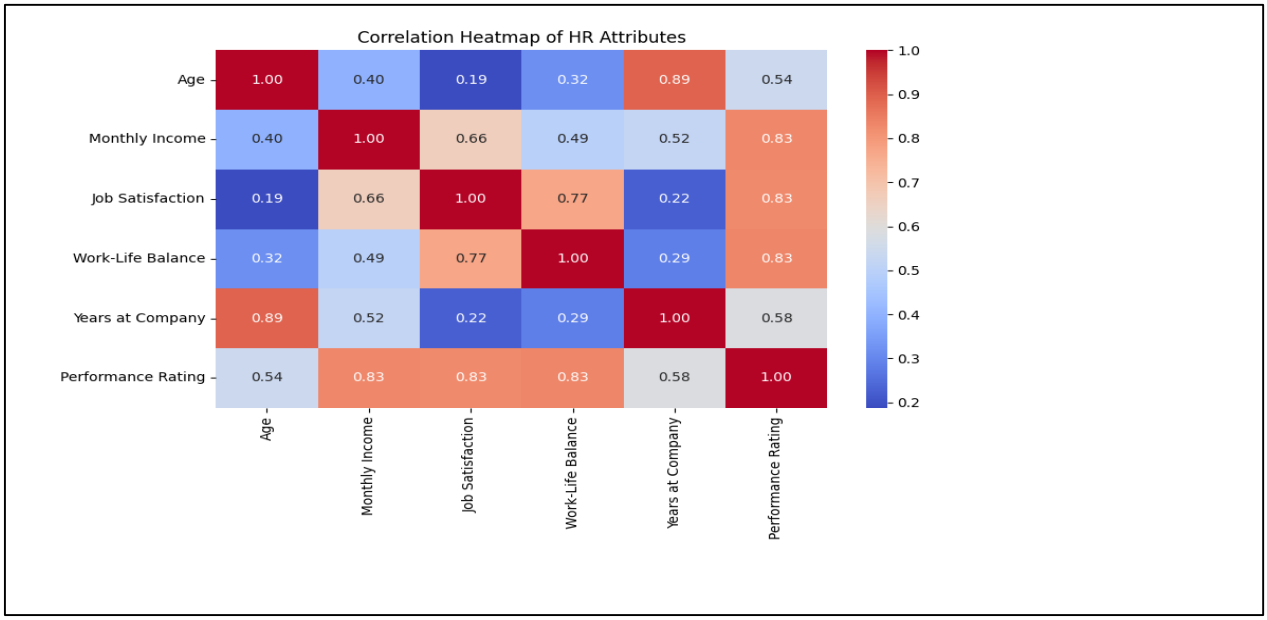
6.3 Numeric Variable Relationships

- The distributions and correlations among suitable numeric variables can be described in this section.
- The exact coefficients should be included only when they have been calculated using the actual dataset.

6.4 Project Output

- The project outputs consist of the inspected or cleaned employee dataset, analysis notebook, summary tables, visualizations, and project report.
- Only the files and charts that have actually been created and checked should be listed.

6.5 Interpretation



CHAPTER 7: KEY FINDINGS AND RECOMMENDATIONS

7.1 Key Findings

- Employee Turnover: The dataset can be used to understand the number of employees who have left the organization compared with those who continue working.
- Department-Wise Analysis: Comparing different departments helps identify variations in employee attrition across organizational units.
- Job Satisfaction: Employee satisfaction levels can be examined to understand their possible relationship with attrition.
- Salary and Experience: Monthly income and years of experience at the company can be analyzed to identify patterns associated with employee retention.
- Work Environment: Factors such as overtime and job role can be explored to understand their possible association with employee turnover.
- Data-Driven Decisions: The analysis provides useful insights that can help organizations improve employee satisfaction and develop better retention strategies.

7.2 Recommendations

- Use a reliable, well-documented dataset with clearly defined attributes and category labels.
- Document missing-value handling, duplicate checks, and all preprocessing decisions.
- Use attrition rates as well as counts when comparing groups, and show group sizes.
- Avoid making causal claims from charts or correlations alone.
- Protect employee privacy and avoid including directly identifying information in reports or public notebooks.
- If a predictive model is developed later, define the target, separate training and test data, and evaluate it using suitable metrics such as precision, recall, F1-score, and a confusion matrix.

CHAPTER 8: LIMITATIONS AND FUTURE ENHANCEMENTS

8.1 Limitations

- Findings depend on the size, quality, representativeness, and source of the selected dataset.
- Missing or inconsistently defined fields can affect comparisons and interpretation.
- Observed associations do not prove that a particular factor causes employee attrition.
- Results from one dataset may not generalize to other organizations, roles, or time periods.
- This EDA report does not demonstrate a trained and evaluated machine-learning prediction model unless such work is completed and documented separately.

8.2 Future Enhancements

- Use a larger, relevant, and ethically sourced dataset with documented field definitions.
- Add time-based analysis if reliable employee and exit dates are available.
- Improve data validation and document preprocessing decisions.
- If sufficient data and a justified use case are available, train and compare suitable classification models with a leakage-aware evaluation process.
- Review fairness, privacy, and the implications of any future model before practical use.

CHAPTER 9: CONCLUSION

- This mini project outlines an exploratory data analysis workflow for Employee Attrition Analysis using Python, Pandas, Matplotlib, and Seaborn.
- The workflow includes data inspection, preprocessing, descriptive statistics, and visualization of employee-related attributes. The completed analysis can help describe how attrition is distributed and how it varies across the groups represented in the dataset.
- All numerical findings must be based on the actual dataset and generated charts. The results are exploratory, do not establish causation, and should not be used alone to make decisions about individual employees.
- Further work would require reliable data, documented definitions, careful evaluation, and appropriate privacy safeguards.

REFERENCES

- The employee attrition dataset selected by the project team: [employee_attrition_messy_dataset.xlsx, employee_attrition_cleaned_dataset.xlsx].
- Pandas Documentation: https://pandas.pydata.org/docs/
- Matplotlib Documentation: https://matplotlib.org/stable/
- Seaborn Documentation: https://seaborn.pydata.org/
- Google Colab Documentation: https://colab.research.google.com/

APPENDIX: PROJECT FILES AND VISUALIZATIONS

The project materials may include the following files, depending on what has actually been prepared

- Employee Attrition Messy Dataset – [employee_attrition_messy_dataset.xlsx]
- Cleaned Employee Dataset – [employee_attrition_cleaned_dataset.xlsx]
- EDA Notebook – [Mini project.pdf]
- Figure 1: Attrition Distribution [bar chart.png]
- Figure 2: Attrition by Department or Job Role, if fields are available [scatter plot.png]
- Figure 3: Age or Years-at-Company Distribution [correlation heat map.png]

GITHUB LINK:

https://github.com/thrishasakthivel975-cloud/Employee_Attrition_Analysis/tree/main